

An Examination of Text Shaking Correction Methods for AR Walking

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ABSTRACT

One problem with walking in AR is less readability of displayed text. Head shaking causes the displayed text to shake. The screen coordinate system(SCS) or world coordinate system(WCS) is used for displaying text with different effective distances. We propose methods to correct text shaking by combining SCS and WCS.

KEYWORDS

Augmented reality, Text shaking, Coordinate systems

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1 INTRODUCTION

In recent years, augmented reality (AR) using AR glasses and head mounted displays (HMDs) has been attracting attention. AR is a technology that superimposes visual information on the real space, and it is expected to be used in our daily lives. However, the screen and displayed text are shaking when we are walking with AR glasses or HMDs.

Two text display methods exist: the screen coordinate system (SCS) and the world coordinate system (WCS), as shown in Figure 1. Related study [Fukushima et al. 2020] has shown that the effective distance of readability in SCS and WCS was different. The distance 0.80 m in front of the eye at rest is called the head fixation point (HFP) [Hirasaki 2000], at which the gaze is stable. When a text is displayed forward less than 0.80 m, SCS that follows the head is considered suitable [Fukushima et al. 2020]. On the other hand, when the text is displayed further than 0.80 m, WCS anchored in the real world is appropriate [Fukushima et al. 2020]. Moreover, there is a reflection called the vestibulo-ocular reflex (VOR) [Broussard et al. 2010]. The VOR can bring on visual stability caused by our learning of the text shaking while we walk. When the head moves up and down while walking, unconscious movement of the eyes occurs to stabilize the image on the retina. Therefore, WCS is presumed to be superior at all distances. However, it is not clear which is better,



Figure 1: Left: Initial position of the head and displayed text, Center: SCS after moving the head, Right: WCS after moving the head

SCS or WCS. In addition to these visual properties, as regards WCS, when we move with AR glasses or HMDs, the text does not follow.

In this study, we propose two methods to incorporate WCS position coordinates into SCS so that the text can follow. Then, we investigate the effects of SCS and our proposed methods on the text readability, comfort, reading speed (character per minute (CPM)) and eye movement. We confirm the advantages of our methods as regards the correction of the text shaking.

2 EXPERIMENT

The purpose of this experiment is to determine the effectiveness of our methods on the text readability, comfort, CPM and eye movement. In this experiment, we conduct a subject experiment in which we change the distances of the text displayed on the HMD and compare SCS and our methods. In our methods, we incorporate WCS position coordinates into SCS in two ways, which we call WCS:Y and WCS:XY.

2.1 Display Methods and Text Distances

Three types of text display methods were used. The first was SCS, in which the text was displayed a fixed position on the screen even if the head moved up and down, left and right, back and forth (including the head rotation). The second was WCS:Y. In WCS:Y, SCS was used, but when the head moved up and down, the same processing as WCS was performed only in the up and down direction (Y-axis direction). Therefore, when the head moved left and right, back and forth, the text was displayed in a fixed position on the screen as same as SCS. On the other hand, when the head moved up and down, the text was fixed at a point in the real world as same as WCS. In other words, when the head moved up, the position of the text was lowered relative to the head. The third was WCS:XY. In WCS:XY, SCS was used, but when the head moved up

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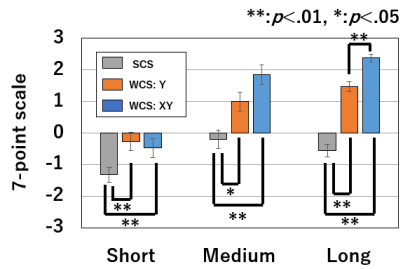


Figure 2: Results of the text readability

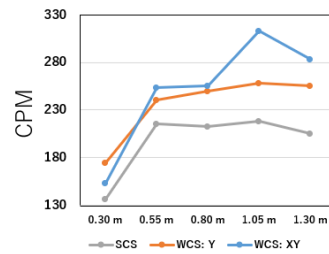


Figure 3: Results of the CPM

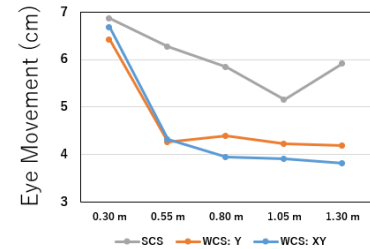


Figure 4: Results of the eye movement

and down, left and right (including the head rotation), the same processing as WCS was performed in the up and down, left and right directions (X and Y-axes directions). Therefore, SCS was used only when the head moved back and forth.

The number of characters to be displayed was set at thirty (three lines of ten characters each). They included two types of Japanese characters "hiragana" and "kanji" and made a sentence. We prepared 95 different sentences and randomly presented them. The subject read the sentence from the upper left to the lower right. The font was MS Gothic, which is a standard font in Windows. The text color was black with the white outline. The outline maintained the visibility from the black background.

Five text distances (0.30, 0.55, 0.80, 1.05, 1.30 m) were used. In all the five distances, the viewing angle was adjusted to 1.53 degrees, so the size of the visible font was same.

2.2 Procedure

In the experiment, the subject rode on a treadmill (4 km/h) while wearing an HMD(HTC, VIVE Pro Eye). After the practice trial, the subject performed the main trial of steps (1) through (3) below:

- (1) When the subject is walking, click the button on the right hand at any time to display the text.
- (2) Read the text as quickly and accurately as possible.
- (3) Evaluate "Did you feel that the text was easy to read?" on a 7-point scale (disagree/agree (very strongly (-3)/(+3), strongly (-2)/(+2), slightly (-1)/(+1)), neutral (0)).

To obtain quantitative data, we measured the text reading speed (CPM) and the eye movement.

The main trials were performed under all fifteen conditions: three different display methods and five different text distances. The order of the conditions was randomized for each subject. Twenty subjects (one female and nineteen male university students) were participated in the experiment.

3 RESULTS AND DISCUSSION

The total of 300 data points was analyzed (fifteen conditions x twenty subjects). The display methods were compared in the following distances: 0.30 and 0.55 m for short distances, 0.80 m for medium distance, and 1.05 and 1.30 m for long distances. Figure 2 shows the results of the readability, indicating the mean and standard error for the display methods (SCS, WCS:Y and WCS:XY). For all distances, the three methods showed significant differences in the readability using a one-way repeated measures ANOVA ($F(2,117)=9.53$ (short), $F(2,57)=15.8$ (medium), $F(2,117)=110.9$ (long), $p<.01$,

respectively). WCS:Y and WCS:XY improved the readability from SCS (Holm, $p<.05$, $p<.01$, respectively). Furthermore, for the long distances, WCS:XY was significantly superior to the others (Holm, $p<.01$, respectively). Therefore, the results differ from the related study [Fukushima et al. 2020] that for distances shorter than 0.8 m, using SCS rather than WCS increases the readability. In addition, WCS:XY is better than WCS:Y, which means that WCS is performed more effective both in the vertical and horizontal directions than only in the vertical direction.

Figure 3 shows the average CPM for each of the three display methods at each distance. For all display methods, the CPM increased as the text distance increased with the highest CPM at 1.05 m and then it decreased at 1.30 m. The results showed that the text distance at around 1.05 m was the best for rapid reading of the text.

In Figure 4, the average of the eye movement for each of the three display methods at each distance is shown. At 0.30 m, the movement was the largest for all the display methods, and it decreased as the distance increased. In particular, WCS:Y and WCS:XY showed that the movement decreased to about 4.0 cm at the distances greater than 0.30 m. On the other hand, SCS was shakier than WCS:Y and WCS:XY. Therefore, with our methods that incorporates WCS with SCS, the eye movement is expected in less shaking and thus a more comfortable reading can be experienced.

In the above comparison of the display methods in terms of the text readability, the CPM and the eye movement, WCS:Y and WCS:XY was superior to SCS. In contrast to the related study [Fukushima et al. 2020], WCS was superior at distances shorter than 0.80 m. Therefore, for distances shorter than 0.80 m, the effect of VOR correction is stronger than the effect of HFP, suggesting that WCS is superior to SCS.

4 CONCLUSION

In this study, we investigated the text shake correction methods for AR walking. From the experiment results, we found that display methods that incorporate the position information of WCS into SCS enable better readability, more comfort and faster text reading at text distances of 0.55 m or more.

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